Neural Computation -- 1995

80 article(s), all issues

1. Cascade Error Projection Learning Theory

Tuan A. Duong

(no DOI)

The cascade correlation based neural network learning algorithm has drawn a lot of attention.

---

2. Optimal Bounds of Learning Curves in Concept Learning

H. Takahashi

(no DOI)

No abstract available.

---

3. On Neural Circuits and Cognition

Michael S. Gazzaniga

https://doi.org/10.1162/neco.1995.7.1.1 | pp. 1-12

January 01 1995 On Neural Circuits and Cognition In Special Collection: CogNet Michael S. Gazzaniga Michael S. Gazzaniga Center for Neuroscience, University of California at Davis, Davis, CA 95616 USA Search for other works by this author on: This Site Google Scholar Author and Article Information Michael S. Gazzaniga Center for Neuroscience, University of California at Davis, Davis, CA 95616 USA Received: September 01 1993 Accepted: May 16 1994 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1995 Massachusetts Institute of Technology1995 Neural Computation (1995) 7 (1): 1–12. https://doi.org/10.1162/neco.1995.7.1.1 Article history Received: September 01 1993 Accepted: May 16 1994 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn Email Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Michael S. Gazzaniga; On Neural Circuits and Cognition. Neural Comput 1995; 7 (1): 1–12. doi: https://doi.org/10.1162/neco.1995.7.1.1 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1995 Massachusetts Institute of Technology1995 Article PDF first page preview Close Modal You do not currently have access to this content.

---

4. The EM Algorithm and Information Geometry in Neural Network Learning

■■■-■■■ ■■■■■

https://doi.org/10.1162/neco.1995.7.1.13 | pp. 13-18

Hidden units play an important role in neural networks, although their activation values are unknown in many learning situations. The EM algorithm (statistical algorithm) and the em algorithm (information-geometric one) have been proposed so far in this connection, and the effectiveness of such algorithms is recognized in many areas of research. The present note points out that these two algorithms are equivalent under a certain condition, although they are different in general.

---

5. Convergence Theorems for Hybrid Learning Rules

Michel Bena■■m

https://doi.org/10.1162/neco.1995.7.1.19 | pp. 19-24

January 01 1995 Convergence Theorems for Hybrid Learning Rules In Special Collection: CogNet Michel Benaim Michel Benaim Department of Mathematics, University of California at Berkeley, Berkeley, CA 94720 USA Search for other works by this author on: This Site Google Scholar Author and Article Information Michel Benaim Department of Mathematics, University of California at Berkeley, Berkeley, CA 94720 USA Received: October 28 1993 Accepted: May 20 1994 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1995 Massachusetts Institute of Technology1995 Neural Computation (1995) 7 (1): 19–24. https://doi.org/10.1162/neco.1995.7.1.19 Article history Received: October 28 1993 Accepted: May 20 1994 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn Email Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Michel Benaim; Convergence Theorems for Hybrid Learning Rules. Neural Comput 1995; 7 (1): 19–24. doi: https://doi.org/10.1162/neco.1995.7.1.19 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1995 Massachusetts Institute of Technology1995 Article PDF first page preview Close Modal You do not currently have access to this content.

---

6. A Type of Duality between Self-Organizing Maps and Minimal Wiring

Graeme Mitchison

https://doi.org/10.1162/neco.1995.7.1.25 | pp. 25-35

I show here that two interpretations of neural maps are closely related. The first, due to Kohonen, sees these maps as forming by an adaptive process in response to stimuli. The second—the minimal wiring or dimension-reduction perspective—interprets the maps as the solution of a minimization problem, where the goal is to keep the “wiring” between neurons with similar receptive fields as short as possible. Recent work by Luttrell provides a bridging concept, by showing that Kohonen's algorithm can be regarded as an approximation to gradient descent on a certain functional. I show how this functional can be generalized in a way that allows it to be interpreted as a measure of wirelength.

---

7. Development of Oriented Ocular Dominance Bands as a Consequence of Areal Geometry

H.-U. Bauer

https://doi.org/10.1162/neco.1995.7.1.36 | pp. 36-50

It has been hypothesized that the different appearance of ocular dominance bands in the cat and the monkey is a consequence of the different mapping geometries in these species (LeVay et al. 1985; Anderson et al. 1988). Here I investigate the impact of areal geometries on the preferred direction of ocular dominance bands in two adaptive map formation models, the self-organizing feature map and the elastic net algorithm. In the case of the self-organizing feature map, the occurrence of instabilities that correspond to ocular dominance bands can be analytically investigated. The instabilities automatically yield stripes of correct orientation. These analytic results are complemented by simulations. In the case of the elastic net algorithm, simulations reveal two different parameter regimes of the algorithm, only one of which leads to stripes of correct orientation. The results suggest that neighborhood preservation in visual maps is enforced in the backward direction, such that neighboring cells in the cortex have neighboring receptive fields, and not vice versa.

---

8. A Multiple Cause Mixture Model for Unsupervised Learning

Eric Saund

https://doi.org/10.1162/neco.1995.7.1.51 | pp. 51-71

This paper presents a formulation for unsupervised learning of clusters reflecting multiple causal structure in binary data. Unlike the “hard” k-means clustering algorithm and the “soft” mixture model, each of which assumes that a single hidden event generates each data point, a multiple cause model accounts for observed data by combining assertions from many hidden causes, each of which can pertain to varying degree to any subset of the observable dimensions. We employ an objective function and iterative gradient descent learning algorithm resembling the conventional mixture model. A crucial issue is the mixing function for combining beliefs from different cluster centers in order to generate data predictions whose errors are minimized both during recognition and learning. The mixing function constitutes a prior assumption about underlying structural regularities of the data domain; we demonstrate a weakness inherent to the popular weighted sum followed by sigmoid squashing, and offer alternative forms of the nonlinearity for two types of data domain. Results are presented demonstrating the algorithm's ability successfully to discover coherent multiple causal representations in several experimental data sets.

---

9. Similarity Metric Learning for a Variable-Kernel Classifier

David Lowe

https://doi.org/10.1162/neco.1995.7.1.72 | pp. 72-85

Nearest-neighbor interpolation algorithms have many useful properties for applications to learning, but they often exhibit poor generalization. In this paper, it is shown that much better generalization can be obtained by using a variable interpolation kernel in combination with conjugate gradient optimization of the similarity metric and kernel size. The resulting method is called variable-kernel similarity metric (VSM) learning. It has been tested on several standard classification data sets, and on these problems it shows better generalization than backpropagation and most other learning methods. The number of parameters that must be determined through optimization are orders of magnitude less than for backpropagation or radial basis function (RBF) networks, which may indicate that the method better captures the essential degrees of

variation in learning. Other features of VSM learning are discussed that make it relevant to models for biological learning in the brain.

---

10. Unsupervised Mutual Information Criterion for Elimination of Overtraining in Supervised Multilayer Networks

Gustavo Deco, William Finnoff, H. G. Zimmermann

https://doi.org/10.1162/neco.1995.7.1.86 | pp. 86-107

Controlling the network complexity in order to prevent overfitting is one of the major problems encountered when using neural network models to extract the structure from small data sets. In this paper we present a network architecture designed for use with a cost function that includes a novel complexity penalty term. In this architecture the outputs of the hidden units are strictly positive and sum to one, and their outputs are defined as the probability that the actual input belongs to a certain class formed during learning. The penalty term expresses the mutual information between the inputs and the extracted classes. This measure effectively describes the network complexity with respect to the given data in an unsupervised fashion. The efficiency of this architecture/penalty-term when combined with backpropagation training, is demonstrated on a real world economic time series forecasting problem. The model was also applied to the benchmark sunspot data and to a synthetic data set from the statistics community.

---

11. Training with Noise is Equivalent to Tikhonov Regularization

Chris Bishop

https://doi.org/10.1162/neco.1995.7.1.108 | pp. 108-116

It is well known that the addition of noise to the input data of a neural network during training can, in some circumstances, lead to significant improvements in generalization performance. Previous work has shown that such training with noise is equivalent to a form of regularization in which an extra term is added to the error function. However, the regularization term, which involves second derivatives of the error function, is not bounded below, and so can lead to difficulties if used directly in a learning algorithm based on error minimization. In this paper we show that for the purposes of network training, the regularization term can be reduced to a positive semi-definite form that involves only first derivatives of the network mapping. For a sum-of-squares error function, the regularization term belongs to the class of generalized Tikhonov regularizers. Direct minimization of the regularized error function provides a practical alternative to training with noise.

---

12. Bayesian Regularization and Pruning Using a Laplace Prior

Peter M. Williams

https://doi.org/10.1162/neco.1995.7.1.117 | pp. 117-143

Standard techniques for improved generalization from neural networks include weight decay and pruning. Weight decay has a Bayesian interpretation with the decay function corresponding to a prior over weights. The method of transformation groups and maximum entropy suggests a Laplace rather than a gaussian prior. After training, the weights then arrange themselves into two classes: (1) those with a common

sensitivity to the data error and (2) those failing to achieve this sensitivity and that therefore vanish. Since the critical value is determined adaptively during training, pruning—in the sense of setting weights to exact zeros—becomes an automatic consequence of regularization alone. The count of free parameters is also reduced automatically as weights are pruned. A comparison is made with results of MacKay using the evidence framework and a gaussian regularizer.

---

13. Empirical Risk Minimization versus Maximum-Likelihood Estimation: A Case Study

Ronny Meir

https://doi.org/10.1162/neco.1995.7.1.144 | pp. 144-157

We study the interaction between input distributions, learning algorithms, and finite sample sizes in the case of learning classification tasks. Focusing on the case of normal input distributions, we use statistical mechanics techniques to calculate the empirical and expected (or generalization) errors for several well-known algorithms learning the weights of a single-layer perceptron. In the case of spherically symmetric distributions within each class we find that the simple Hebb rule, corresponding to maximum-likelihood parameter estimation, outperforms the other more complex algorithms, based on error minimization. Moreover, we show that in the regime where the overlap between the classes is large, algorithms with low empirical error do worse in terms of generalization, a phenomenon known as overtraining.

---

14. Learning a Decision Boundary from Stochastic Examples: Incremental Algorithms with and without Queries

Yoshiyuki Kabashima, Shigeru Shinomoto

https://doi.org/10.1162/neco.1995.7.1.158 | pp. 158-172

Even if it is not possible to reproduce a target input-output relation, a learning machine should be able to minimize the probability of making errors. A practical learning algorithm should also be simple enough to go without memorizing example data, if possible. Incremental algorithms such as error backpropagation satisfy this requirement. We propose incremental algorithms that provide fast convergence of the machine parameter θ to its optimal choice θ o with respect to the number of examples t. We will consider the binary choice model whose target relation has a blurred boundary and the machine whose parameter θ specifies a decision boundary to make the output prediction. The question we wish to address here is how fast θ can approach θ o , depending upon whether in the learning stage the machine can specify inputs as queries to the target relation, or the inputs are drawn from a certain distribution. If queries are permitted, the machine can achieve the fastest convergence, $(\theta - \theta \text{ o }) 2 \sim O(t -1)$. If not, $O(t -1)$ convergence is generally not attainable. For learning without queries, we showed in a previous paper that the error minimum algorithm exhibits a slow convergence $(\theta - \theta \text{ o }) 2 \sim O(t -2/3)$. We propose here a practical algorithm that provides a rather fast convergence, $O(t -4/5)$. It is possible to further accelerate the convergence by using more elaborate algorithms. The fastest convergence turned out to be $O[(\ln t) 2 t -1]$. This scaling is considered optimal among possible algorithms, and is not due to the incremental nature of our algorithm.

---

15. Arithmetic Perceptrons

Sergio A. Cannas

https://doi.org/10.1162/neco.1995.7.1.173 | pp. 173-181

A feedforward layered neural network (perceptron) with one hidden layer, which adds two N-bit binary numbers is constructed. The set of synaptic strengths and thresholds is obtained exactly for different architectures of the network and for arbitrary N. These structures can be easily generalized to perform more complicated arithmetic operations (like subtraction).

---

16. Compensatory Mechanisms in an Attractor Neural Network Model of Schizophrenia

D. Horn, Eytan Ruppin

https://doi.org/10.1162/neco.1995.7.1.182 | pp. 182-205

We investigate the effect of synaptic compensation on the dynamic behavior of an attractor neural network receiving its input stimuli as external fields projecting on the network. It is shown how, in the face of weakened inputs, memory performance may be preserved by strengthening internal synaptic connections and increasing the noise level. Yet, these compensatory changes necessarily have adverse side effects, leading to spontaneous, stimulus-independent retrieval of stored patterns. These results can support Stevens' recent hypothesis that the onset of schizophrenia is associated with frontal synaptic regeneration, occurring subsequent to the degeneration of temporal neurons projecting on these areas.

---

17. Real-Time Control of a Tokamak Plasma Using Neural Networks

Chris Bishop, Paul S. Haynes, Mike Smith, Tom N. Todd, D. L. Trotman

https://doi.org/10.1162/neco.1995.7.1.206 | pp. 206-217

In this paper we present results from the first use of neural networks for real-time control of the high-temperature plasma in a tokamak fusion experiment. The tokamak is currently the principal experimental device for research into the magnetic confinement approach to controlled fusion. In an effort to improve the energy confinement properties of the high-temperature plasma inside tokamaks, recent experiments have focused on the use of noncircular cross-sectional plasma shapes. However, the accurate generation of such plasmas represents a demanding problem involving simultaneous control of several parameters on a time scale as short as a few tens of microseconds. Application of neural networks to this problem requires fast hardware, for which we have developed a fully parallel custom implementation of a multilayer perceptron, based on a hybrid of digital and analogue techniques.

---

18. Regularization Theory and Neural Networks Architectures

Federico Girosi, Michael Jones, Tomaso Poggio

https://doi.org/10.1162/neco.1995.7.2.219 | pp. 219-269

We had previously shown that regularization principles lead to approximation schemes that are equivalent to networks with one layer of hidden units, called regularization networks. In particular, standard smoothness

functionals lead to a subclass of regularization networks, the well known radial basis functions approximation schemes. This paper shows that regularization networks encompass a much broader range of approximation schemes, including many of the popular general additive models and some of the neural networks. In particular, we introduce new classes of smoothness functionals that lead to different classes of basis functions. Additive splines as well as some tensor product splines can be obtained from appropriate classes of smoothness functionals. Furthermore, the same generalization that extends radial basis functions (RBF) to hyper basis functions (HBF) also leads from additive models to ridge approximation models, containing as special cases Breiman's hinge functions, some forms of projection pursuit regression, and several types of neural networks. We propose to use the term generalized regularization networks for this broad class of approximation schemes that follow from an extension of regularization. In the probabilistic interpretation of regularization, the different classes of basis functions correspond to different classes of prior probabilities on the approximating function spaces, and therefore to different types of smoothness assumptions. In summary, different multilayer networks with one hidden layer, which we collectively call generalized regularization networks, correspond to different classes of priors and associated smoothness functionals in a classical regularization principle. Three broad classes are (1) radial basis functions that can be generalized to hyper basis functions, (2) some tensor product splines, and (3) additive splines that can be generalized to schemes of the type of ridge approximation, hinge functions, and several perceptron-like neural networks with one hidden layer.

---

19. A Counterexample to Temporal Differences Learning

Dimitri P. Bertsekas

https://doi.org/10.1162/neco.1995.7.2.270 | pp. 270-279

Sutton's TD(λ) method aims to provide a representation of the cost function in an absorbing Markov chain with transition costs. A simple example is given where the representation obtained depends on λ. For λ = 1 the representation is optimal with respect to a least-squares error criterion, but as λ decreases toward 0 the representation becomes progressively worse and, in some cases, very poor. The example suggests a need to understand better the circumstances under which TD(0) and Q-learning obtain satisfactory neural network-based compact representations of the cost function. A variation of TD(0) is also given, which performs better on the example.

---

20. New Perceptron Model Using Random Bitstreams

Eel-Wan Lee, Soo■Ik Chae

https://doi.org/10.1162/neco.1995.7.2.280 | pp. 280-283

A very high precision is needed to implement the adder using stochastic computation (or pulse arithmetic) in modern VLSI technology. In this paper we propose a new model of perceptron using random bitstreams that alleviates this problem.

---

21. On the Ordering Conditions for Self-Organizing Maps

Marco Budinich, John G. Taylor

https://doi.org/10.1162/neco.1995.7.2.284 | pp. 284-289

We present a geometric interpretation of ordering in self-organizing feature maps. This view provides simpler proofs of Kohonen ordering theorem and of convergence to an ordered state in the one-dimensional case. At the same time it explains intuitively the origin of the problems in higher dimensional cases. Furthermore it provides a geometric view of the known characteristics of learning in self-organizing nets.

---

22. A Simple Competitive Account of Some Response Properties of Visual Neurons in Area MSTd

Ruye Wang

https://doi.org/10.1162/neco.1995.7.2.290 | pp. 290-306

A simple and biologically plausible model is proposed to simulate the optic flow computation taking place in the dorsal part of medial superior temporal (MSTd) area of the visual cortex in the primates' brain. The model is a neural network composed of competitive learning layers. The input layer of the network simulates the neurons in the middle temporal (MT) area that selectively respond to the visual stimuli of the input motion patterns with different local velocities. The output layer of the network simulates the MSTd neurons that selectively respond to different types of optic flow motion patterns including planar, circular, radial, and spiral motions. Simulation results obtained from this model show that the behaviors of the output nodes of the network resemble very closely the known responsive properties of the MSTd neurons found neurophysiologically, such as the existence of three types of MSTd neurons that respond, respectively, to one, two, or three types of the input motion patterns with different position dependences, and the continuum of response selectivity formed by the three types of neurons.

---

23. Synchrony in Excitatory Neural Networks

D. Hansel, Germán Mato, C. Meunier

https://doi.org/10.1162/neco.1995.7.2.307 | pp. 307-337

Synchronization properties of fully connected networks of identical oscillatory neurons are studied, assuming purely excitatory interactions. We analyze their dependence on the time course of the synaptic interaction and on the response of the neurons to small depolarizations. Two types of responses are distinguished. In the first type, neurons always respond to small depolarization by advancing the next spike. In the second type, an excitatory postsynaptic potential (EPSP) received after the refractory period delays the firing of the next spike, while an EPSP received at a later time advances the firing. For these two types of responses we derive general conditions under which excitation destabilizes in-phase synchrony. We show that excitation is generally desynchronizing for neurons with a response of type I but can be synchronizing for responses of type II when the synaptic interactions are fast. These results are illustrated on three models of neurons: the Lapicque integrate-and-fire model, the model of Connor et al., and the Hodgkin-Huxley model. The latter exhibits a type II response, at variance with the first two models, that have type I responses. We then examine the consequences of these results for large networks, focusing on the states of partial coherence that emerge. Finally, we study the Lapicque model and the model of Connor et al. at large coupling and show that excitation can be desynchronizing even beyond the weak coupling regime.

---

24. Decorrelated Hebbian Learning for Clustering and Function Approximation

Gustavo Deco, Dragan Obradovi■

https://doi.org/10.1162/neco.1995.7.2.338 | pp. 338-348

This paper presents a new learning paradigm that consists of a Hebbian and anti-Hebbian learning. A layer of radial basis functions is adapted in an unsupervised fashion by minimizing a two-element cost function. The first element maximizes the output of each gaussian neuron and it can be seen as an implementation of the traditional Hebbian learning law. The second element of the cost function reinforces the competitive learning by penalizing the correlation between the nodes. Consequently, the second term has an “anti-Hebbian” effect that is learned by the gaussian neurons without the implementation of lateral inhibition synapses. Therefore, the decorrelated Hebbian learning (DHL) performs clustering in the input space avoiding the “nonbiological” winner-take-all rule. In addition to the standard clustering problem, this paper also presents an application of the DHL in function approximation. A scaled piece-wise linear approximation of a function is obtained in the supervised fashion within the local regions of its domain determined by the DHL. For comparison, a standard single hidden-layer gaussian network is optimized with the initial centers corresponding to the DHL. The efficiency of the algorithm is demonstrated on the chaotic Mackey-Glass time series.

---

25. Identification Using Feedforward Networks

Asriel U. Levin, Kumpati S. Narendra

https://doi.org/10.1162/neco.1995.7.2.349 | pp. 349-369

This paper is concerned with the identification of an unknown nonlinear dynamic system when only the inputs and outputs are accessible for measurement. Specifically we investigate the use of feedforward neural networks as models for the input-output behavior of such systems. Relying on the approximation capabilities of feedforward neural networks and under mild assumptions regarding the properties of the underlying nonlinear system, it is shown that there exists a feedforward network that for almost all inputs (an open and dense set) will display the input-output behavior of the system.

---

26. An HMM/MLP Architecture for Sequence Recognition

Sung■Bae Cho, Jin Hee Kim

https://doi.org/10.1162/neco.1995.7.2.358 | pp. 358-369

This paper presents a hybrid architecture of hidden Markov models (HMMs) and a multilayer perceptron (MLP). This exploits the discriminative capability of a neural network classifier while using HMM formalism to capture the dynamics of input patterns. The main purpose is to improve the discriminative power of the HMM-based recognizer by additionally classifying the likelihood values inside them with an MLP classifier. To appreciate the performance of the presented method, we apply it to the recognition problem of on-line handwritten characters. Simulations show that the proposed architecture leads to a significant improvement in generalization performance over conventional approaches to sequential pattern recognition.

---

27. Learning Linear Threshold Approximations Using Perceptrons

Tom Bylander

https://doi.org/10.1162/neco.1995.7.2.370 | pp. 370-379

We demonstrate sufficient conditions for polynomial learnability of suboptimal linear threshold functions using perceptrons. The central result is as follows. Suppose there exists a vector w * , of n weights (including the threshold) with "accuracy" 1 – α, "average error" η, and "balancing separation" σ, i.e., with probability 1 – α, w * correctly classifies an example x; over examples incorrectly classified by w * , the expected value of |w * · x| is η (source of inaccuracy does not matter); and over a certain portion of correctly classified examples, the expected value of |w * · x| is σ. Then, with probability 1 – δ, the perceptron achieves accuracy at least 1 – [■ + α(1 + η/σ)] after O[n■ –2 σ –2 (ln 1/δ)] examples.

---

28. An Algorithm for Building Regularized Piecewise Linear Discrimination Surfaces: The Perceptron Membrane

Guillaume Deffuant

https://doi.org/10.1162/neco.1995.7.2.380 | pp. 380-398

The perceptron membrane is a new connectionist model that aims at solving discrimination (classification) problems with piecewise linear surfaces. The discrimination surfaces of perceptron membranes are defined by the union of convex polyhedrons. Starting from only one convex polyhedron, new facets and new polyhedrons are added during learning. Moreover, the positions and orientations of the facets are continuously adapted according to the training examples. Considering each facet as a perceptron cell, a geometric credit assignment provides a local training domain to each perceptron of the network. This enables one to apply statistical theorems on the probability of good generalization for each unit on its learning domain, and gives a reliable criterion for perceptron elimination (using Vapnik-Chervonenkis dimension). Furthermore, a regularization procedure is implemented. The model efficiency is demonstrated on well-known problems such as the 2-spirals or waveforms.

---

29. The Upward Bias in Measures of Information Derived from Limited Data Samples

Alessandro Treves, Stefano Panzeri

https://doi.org/10.1162/neco.1995.7.2.399 | pp. 399-407

Extracting information measures from limited experimental samples, such as those normally available when using data recorded in vivo from mammalian cortical neurons, is known to be plagued by a systematic error, which tends to bias the estimate upward. We calculate here the average of the bias, under certain conditions, as an asymptotic expansion in the inverse of the size of the data sample. The result agrees with numerical simulations, and is applicable, as an additive correction term, to measurements obtained under such conditions. Moreover, we discuss the implications for measurements obtained through other usual procedures.

---

30. Representation of Similarity in Three-Dimensional Object Discrimination

Shimon Edelman
https://doi.org/10.1162/neco.1995.7.2.408 | pp. 408-423

How does the brain represent visual objects? In simple perceptual generalization tasks, the human visual system performs as if it represents the stimuli in a low-dimensional metric psychological space (Shepard 1987). In theories of three-dimensional (3D) shape recognition, the role of feature-space representations [as opposed to structural (Biederman 1987) or pictorial (Ullman 1989) descriptions] has long been a major point of contention. If shapes are indeed represented as points in a feature space, patterns of perceived similarity among different objects must reflect the structure of this space. The feature space hypothesis can then be tested by presenting subjects with complex parameterized 3D shapes, and by relating the similarities among subjective representations, as revealed in the response data by multidimensional scaling (Shepard 1980), to the objective parameterization of the stimuli. The results of four such tests, accompanied by computational simulations, support the notion that discrimination among 3D objects may rely on a low-dimensional feature space representation, and suggest that this space may be spanned by explicitly encoded class prototypes.

---

31. Models of Orientation and Ocular Dominance Columns in the Visual Cortex: A Critical Comparison

Ed Erwin, Klaus Obermayer, Klaus Schulten
https://doi.org/10.1162/neco.1995.7.3.425 | pp. 425-468

Orientation and ocular dominance maps in the primary visual cortex of mammals are among the most thoroughly investigated of the patterns in the cerebral cortex. A considerable amount of work has been dedicated to unraveling both their detailed structure and the neural mechanisms that underlie their formation and development. Many schemes have been proposed, some of which are in competition. Some models focus on development of receptive fields while others focus on the structure of cortical maps, i.e., the arrangement of receptive field properties across the cortex. Each model used different means to determine its success at reproducing experimental map patterns, often relying principally on visual comparison. Experimental data are becoming available that allow a more careful evaluation of models. In this contribution more than 10 of the most prominent models of cortical map formation and structure are critically evaluated and compared with the most recent experimental findings from macaque striate cortex. Comparisons are based on properties of the predicted or measured cortical map patterns. We introduce several new measures for comparing experimental and model map data that reveal important differences between models. We expect that the use of these measures will improve current models by helping determine parameters to match model maps to experimental data now becoming available from a variety of species. Our study reveals that (1) despite apparent differences, many models are based on similar principles and consequently make similar predictions, (2) several models produce orientation map patterns that are not consistent with the experimental data from macaques, regardless of the plausibility of the models' suggested physiological implementations, and (3) no models have yet fully accounted for both the local and the global relationships between orientation and ocular dominance map patterns.

---

32. How Precise is Neuronal Synchronization?

Peter König, Andreas K. Engel, Pieter R. Roelfsema, Wolf Singer
https://doi.org/10.1162/neco.1995.7.3.469 | pp. 469-485

Recent work suggests that synchronization of neuronal activity could serve to define functionally relevant relationships between spatially distributed cortical neurons. At present, it is not known to what extent this hypothesis is compatible with the widely supported notion of coarse coding, which assumes that features of a stimulus are represented by the graded responses of a population of optimally and suboptimally activated cells. To resolve this issue we investigated the temporal relationship between responses of optimally and suboptimally stimulated neurons in area 17 of cat visual cortex. We find that optimally and suboptimally activated cells can synchronize their responses with a precision of a few milliseconds. However, there are consistent and systematic deviations of the phase relations from zero phase lag. Systematic variation of the orientation of visual stimuli shows that optimally driven neurons tend to lead over suboptimally activated cells. The observed phase lag depends linearly on the stimulus orientation and is, in addition, proportional to the difference between the preferred orientations of the recorded cells. Similar effects occur when testing the influence of the movement direction and the spatial frequency of visual stimuli. These results suggest that binding by synchrony can be used to define assemblies of neurons representing a coarse-coded stimulus. Furthermore, they allow a quantitative test of neuronal network models designed to reproduce physiological results on stimulus-specific synchronization.

---

33. Quantitative Analysis of Electrotonic Structure and Membrane Properties of NMDA-Activated Lamprey Spinal Neurons

C. R. Murphey, Lee E. Moore, James T. Buchanan

https://doi.org/10.1162/neco.1995.7.3.486 | pp. 486-506

Parameter optimization methods were used to quantitatively analyze frequency-domain-voltage-clamp data of NMDA-activated lamprey spinal neurons simultaneously over a wide range of membrane potentials. A neuronal cable model was used to explicitly take into account receptors located on the dendritic trees. The driving point membrane admittance was measured from the cell soma in response to a Fourier synthesized point voltage clamp stimulus. The data were fitted to an equivalent cable model consisting of a single lumped soma compartment coupled resistively to a series of equal dendritic compartments. The model contains voltage-dependent NMDA sensitive (INMDA), slow potassium (IK), and leakage (IL) currents. Both the passive cable properties and the voltage dependence of ion channel kinetics were estimated, including the electronic structure of the cell, the steady-state gating characteristics, and the time constants for particular voltage- and time-dependent ionic conductances. An alternate kinetic formulation was developed that consisted of steady-state values for the gating parameters and their time constants at half-activation values as well as slopes of these parameters at half-activation. This procedure allowed independent restrictions on the magnitude and slope of both the steady-state gating variable and its associated time constant. Quantitative estimates of the voltage-dependent membrane ion conductances and their kinetic parameters were used to solve the nonlinear equations describing dynamic responses. The model accurately predicts current clamp responses and is consistent with experimentally measured TTX-resistant NMDA-induced patterned activity. In summary, an analysis method is developed that provides a pragmatic approach to quantitatively describe a nonlinear neuronal system.

---

34. Reduced Representation by Neural Networks with Restricted Receptive Fields

Marco Idiart, Barry Berk, L. F. Abbott
https://doi.org/10.1162/neco.1995.7.3.507 | pp. 507-517

Model neural networks can perform dimensional reductions of input data sets using correlation-based learning rules to adjust their weights. Simple Hebbian learning rules lead to an optimal reduction at the single unit level but result in highly redundant network representations. More complex rules designed to reduce or remove this redundancy can develop optimal principal component representations, but they are not very compelling from a biological perspective. Neurons in biological networks have restricted receptive fields limiting their access to the input data space. We find that, within this restricted receptive field architecture, simple correlation-based learning rules can produce surprisingly efficient reduced representations. When noise is present, the size of the receptive fields can be optimally tuned to maximize the accuracy of reconstructions of input data from a reduced representation.

---

35. Stochastic Single Neurons

Toru Ohira, Jack D. Cowan
https://doi.org/10.1162/neco.1995.7.3.518 | pp. 518-528

We study the stochastic behavior of a single self-exciting model neuron with additive noise, a system that has bistable stochastic dynamics. We use Langevin and Fokker-Planck equations to obtain analytical expressions for the stationary distribution of activities and for the crossing rate between two stable states. We adjust the parameters in these expressions to fit observed histograms of neural activity, thus obtaining what we call an “effective single neuron” for a given network. We construct an effective single neuron from an activity histogram of a representative hidden neuron in a recurrent learning network. We also compare our result with an effective single neuron previously obtained analytically through the adiabatic elimination approximation.

---

36. Memory Recall by Quasi-Fixed-Point Attractors in Oscillator Neural Networks

Tomoki Fukai, Masatoshi Shiino
https://doi.org/10.1162/neco.1995.7.3.529 | pp. 529-548

It is shown that approximate fixed-point attractors rather than synchronized oscillations can be employed by a wide class of neural networks of oscillators to achieve an associative memory recall. This computational ability of oscillator neural networks is ensured by the fact that reduced dynamic equations for phase variables in general involve two terms that can be respectively responsible for the emergence of synchronization and cessation of oscillations. Thus the cessation occurs in memory retrieval if the corresponding term dominates in the dynamic equations. A bottomless feature of the energy function for such a system makes the retrieval states quasi-fixed points, which admit continual rotating motion to a small portion of oscillators, when an extensive number of memory patterns are embedded. An approximate theory based on the self-consistent signal-to-noise analysis enables one to study the equilibrium properties of the neural network of phase variables with the quasi-fixed-point attractors. As far as the memory retrieval by the quasi-fixed points is concerned, the equilibrium properties including the storage capacity of oscillator neural networks are proved to be similar to those of the Hopfield type neural networks.

37. Learning Population Codes by Minimizing Description Length

Richard S. Zemel, Geoffrey E. Hinton

https://doi.org/10.1162/neco.1995.7.3.549 | pp. 549-564

The minimum description length (MDL) principle can be used to train the hidden units of a neural network to extract a representation that is cheap to describe but nonetheless allows the input to be reconstructed accurately. We show how MDL can be used to develop highly redundant population codes. Each hidden unit has a location in a low-dimensional implicit space. If the hidden unit activities form a bump of a standard shape in this space, they can be cheaply encoded by the center of this bump. So the weights from the input units to the hidden units in an autoencoder are trained to make the activities form a standard bump. The coordinates of the hidden units in the implicit space are also learned, thus allowing flexibility, as the network develops a discontinuous topography when presented with different input classes.

38. Competition and Multiple Cause Models

Peter Dayan, Richard S. Zemel

https://doi.org/10.1162/neco.1995.7.3.565 | pp. 565-579

If different causes can interact on any occasion to generate a set of patterns, then systems modeling the generation have to model the interaction too. We discuss a way of combining multiple causes that is based on the Integrated Segmentation and Recognition architecture of Keeler et al. (1991). It is more cooperative than the scheme embodied in the mixture of experts architecture, which insists that just one cause generate each output, and more competitive than the noisy-or combination function, which was recently suggested by Saund (1994a,b). Simulations confirm its efficacy.

39. Bayesian Self-Organization Driven by Prior Probability Distributions

Alan Yuille, Stelios M. Smirnakis, Lei Xu

https://doi.org/10.1162/neco.1995.7.3.580 | pp. 580-595

Recent work by Becker and Hinton (1992) shows a promising mechanism, based on maximizing mutual information assuming spatial coherence, by which a system can self-organize to learn visual abilities such as binocular stereo. We introduce a more general criterion, based on Bayesian probability theory, and thereby demonstrate a connection to Bayesian theories of visual perception and to other organization principles for early vision (Atick and Redlich 1990). Methods for implementation using variants of stochastic learning are described.

40. Adaptive Voting Rules for k-Nearest Neighbors Classifiers

Riccardo Rovatti, Roberto Ragazzoni, Z.M. Kovacs, R. Guerrieri

https://doi.org/10.1162/neco.1995.7.3.594 | pp. 594-605

A simple form of cooperation between the k-nearest neighbors (NN) approach to classification and the neural-like property of adaptation is explored. A tunable, high level k-nearest neighbors decision rule is

defined that comprehends most previous generalizations of the common majority rule. A learning procedure is developed that applies to this rule and exploits those statistical features that can be induced from the training set. The overall approach is tested on a problem of handwritten character recognition. Experiments show that adaptivity in the decision rule may improve the recognition and rejection capability of standard k-NN classifiers.

---

41. Regularization in the Selection of Radial Basis Function Centers

Mark J. L. Orr

https://doi.org/10.1162/neco.1995.7.3.606 | pp. 606-623

Subset selection and regularization are two well-known techniques that can improve the generalization performance of nonparametric linear regression estimators, such as radial basis function networks. This paper examines regularized forward selection (RFS)—a combination of forward subset selection and zero-order regularization. An efficient implementation of RFS into which either delete-1 or generalized cross-validation can be incorporated and a reestimation formula for the regularization parameter are also discussed. Simulation studies are presented that demonstrate improved generalization performance due to regularization in the forward selection of radial basis function centers.

---

42. Bootstrapping Confidence Intervals for Clinical Input Variable Effects in a Network Trained to Identify the Presence of Acute Myocardial Infarction

William G. Baxt, Halbert White

https://doi.org/10.1162/neco.1995.7.3.624 | pp. 624-638

May 01 1995 Bootstrapping Confidence Intervals for Clinical Input Variable Effects in a Network Trained to Identify the Presence of Acute Myocardial Infarction In Special Collection: CogNet William G. Baxt, William G. Baxt Department of Emergency Medicine and Medicine, University of California, San Diego Medical Center, San Diego, CA USA 92093 *Present address: Department of Emergency Medicine, Hospital of the University of Pennsylvania, 3400 Spruce Street, Philadelphia, PA 19104 USA. Search for other works by this author on: This Site Google Scholar Halbert White Halbert White Department of Economics and Institute for Neural Compilation, University of California, San Diego, San Diego, CA USA 92093 Search for other works by this author on: This Site Google Scholar Author and Article Information William G. Baxt Department of Emergency Medicine and Medicine, University of California, San Diego Medical Center, San Diego, CA USA 92093 Halbert White Department of Economics and Institute for Neural Compilation, University of California, San Diego, San Diego, CA USA 92093 *Present address: Department of Emergency Medicine, Hospital of the University of Pennsylvania, 3400 Spruce Street, Philadelphia, PA 19104 USA. Received: April 21 1994 Accepted: July 22 1994 Online ISSN: 1530-888X Print ISSN: 0899-7667 © 1995 Massachusetts Institute of Technology1995 Neural Computation (1995) 7 (3): 624–638. https://doi.org/10.1162/neco.1995.7.3.624 Article history Received: April 21 1994 Accepted: July 22 1994 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn Email Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation William G. Baxt, Halbert White; Bootstrapping Confidence Intervals for Clinical Input Variable Effects in a Network Trained to Identify the Presence of Acute Myocardial Infarction. Neural Comput 1995; 7 (3): 624–638. doi: https://doi.org/10.1162/neco.1995.7.3.624 Download citation file: Ris (Zotero) Reference Manager EasyBib

Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu toolbar search search input Search input auto suggest filter your search All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1995 Massachusetts Institute of Technology1995 Article PDF first page preview Close Modal You do not currently have access to this content.

43. Hints

Yaser S. Abu■Mostafa

https://doi.org/10.1162/neco.1995.7.4.639 | pp. 639-671

The systematic use of hints in the learning-from-examples paradigm is the subject of this review. Hints are the properties of the target function that are known to use independently of the training examples. The use of hints is tantamount to combining rules and data in learning, and is compatible with different learning models, optimization techniques, and regularization techniques. The hints are represented to the learning process by virtual examples, and the training examples of the target function are treated on equal footing with the rest of the hints. A balance is achieved between the information provided by the different hints through the choice of objective functions and learning schedules. The Adaptive Minimization algorithm achieves this balance by relating the performance on each hint to the overall performance. The application of hints in forecasting the very noisy foreign-exchange markets is illustrated. On the theoretical side, the information value of hints is contrasted to the complexity value and related to the VC dimension.

44. Topology and Geometry of Single Hidden Layer Network, Least Squares Weight Solutions

Frans Coetzee, V.L. Stonick

https://doi.org/10.1162/neco.1995.7.4.672 | pp. 672-705

In this paper the topological and geometric properties of the weight solutions for multilayer perceptron (MLP) networks under the MSE error criterion are characterized. The characterization is obtained by analyzing a homotopy from linear to nonlinear networks in which the hidden node function is slowly transformed from a linear to the final sigmoidal nonlinearity. Two different geometric perspectives for this optimization process are developed. The generic topology of the nonlinear MLP weight solutions is described and related to the geometric interpretations, error surfaces, and homotopy paths, both analytically and using carefully constructed examples. These results illustrate that although the natural homotopy provides a practically valuable heuristic for training, it suffers from a number of theoretical and practical difficulties. The linear system is a bifurcation point of the homotopy equations, and solution paths are therefore generically discontinuous. Bifurcations and infinite solutions further occur for data sets that are not of measure zero. These results weaken the guarantees on global convergence and exhaustive behavior normally associated with homotopy methods. However, the analyses presented provide a clear understanding of the relationship between linear and nonlinear perceptron networks, and thus a firm foundation for development of more powerful training methods. The geometric perspectives and generic topological results describing the nature of the solutions are further generally applicable to network analysis and algorithm evaluation.

45. Time-Skew Hebb Rule in a Nonisopotential Neuron

Barak A. Pearlmutter

https://doi.org/10.1162/neco.1995.7.4.706 | pp. 706-712

In an isopotential neuron with rapid response, it has been shown that the receptive fields formed by Hebbian synaptic modulation depend on the principal eigenspace of Q(0), the input autocorrelation matrix, where Qij(tau) = <xi i(t) xi j(t-tau)> and xi i(t) is the input to synapse i at time t (Oja 1982). We relax the assumption of isopotentiality, introduce a time-skewed Hebb rule, and find that the dynamics of synaptic evolution are determined by the principal eigenspace of Q. This matrix is defined by Qij = integral of 0 infinity (Qij * psi i) (tau) Kij (tau) d tau, where Kij (tau) is the neuron's voltage response to a unit current injection at synapse j as measured tau seconds later at synapse i, and psi(tau) is the time course of the opportunity for modulation of synapse i following the arrival of a presynaptic action potential.

---

46. Synapse Models for Neural Networks: From Ion Channel Kinetics to Multiplicative Coefficient w_{ij}

François Chapeau■Blondeau, Nicolas Chambet

https://doi.org/10.1162/neco.1995.7.4.713 | pp. 713-734

This paper relates different levels at which the modeling of synaptic transmission can be grounded in neural networks: the level of ion channel kinetics, the level of synaptic conductance dynamics, and the level of a scalar synaptic coefficient. The important assumptions to reduce a synapse model from one level to the next are explicitly exhibited. This coherent progression provides control on what is discarded and what is retained in the modeling process, and is useful to appreciate the significance and limitations of the resulting neural networks. This methodic simplification terminates with a scalar synaptic efficacy as it is very often used in neural networks, but here its conditions of validity are explicitly displayed. This scalar synapse also comes with an expression that directly relates it to basic quantities of synaptic functioning, and it can be endowed with meaningful physical units and realistic numerical values. In addition, it is shown that the scalar synapse does not receive the same expression in neural networks operating with spikes or with firing rates. These coherent modeling elements can help to improve, adjust, and refine the investigation of neural systems and their remarkable collective properties for information processing.

---

47. Generalization and Analysis of the Lisberger-Sejnowski VOR Model

Ning Qian

https://doi.org/10.1162/neco.1995.7.4.735 | pp. 735-752

Lisberger and Sejnowski (1992) recently proposed a computational model for motor learning in the vestibular-ocular reflex (VOR) system. They showed that the steady-state gain of the system can be modified by changing the ratio of the two time constants along the feedforward and the feedback projections to the Purkinje cell unit in their model VOR network. Here we generalize their model by including two additional time constant variables and two synaptic weight variables, which were set to fixed values in their original model. We derive the stability conditions of the generalized system and thoroughly analyze its steady-state and transient behavior. It is found that the generalized system can display a continuum of behavior with the Lisberger-Sejnowski model and a static model proposed by Miles et al. (1980b) as special

cases. Moreover, although mathematically the Lisberger-Sejnowski model requires two precise relationships among its parameters, the model is robust against small perturbations from the physiological point of view. Additional considerations on the gain of smooth pursuit eye movement, which is believed to share the positive feedback loop with the VOR network, suggest that the VOR network should operate in the parameter range favoring the behavior studied by Lisberger and Sejnowski. Under this condition, the steady-state gain of the VOR is found to depend on all four time constants in the network. The time constant of the Purkinje cell unit should be relatively small in order to achieve effective VOR learning through the modifications of the other time constants. Our analysis provides a thorough characterization of the system and could thus be useful for guiding further physiological tests of the model.

---

48. Stable Adaptive Control of Robot Manipulators Using “Neural” Networks

Robert M. Sanner, Jean-Jacques Slotine

https://doi.org/10.1162/neco.1995.7.4.753 | pp. 753-790

The rapid development and formalization of adaptive signal processing algorithms loosely inspired by biological models can be potentially harnessed for use in flexible new learning control algorithms for nonlinear dynamic systems. However, if such controller designs are to be viable in practice, their stability must be guaranteed and their performance quantified. In this paper, the stable adaptive tracking control designs employing “neural” networks, initially presented in Sanner and Slotine (1992), are extended to classes of multivariable mechanical systems, including robot manipulators, and bounds are developed for the magnitude of the asymptotic tracking errors and the rate of convergence to these bounds. This new algorithm permits simultaneous learning and control, without recourse to an initial identification stage, and is distinguished from previous stable adaptive robotic controllers, e.g. (Slotine and Li 1987), by the relative lack of structure assumed in the design of the control law. The required control is simply considered to contain unknown functions of the measured state variables, and adaptive “neural” networks are used to stably determine, in real time, the entire required functional dependence. While computationally more complex than explicitly model-based techniques, the methods developed in this paper may be effectively applied to the control of many physical systems for which the state dependence of the dynamics is reasonably well understood, but the exact functional form of this dependence, or part thereof, is not, such as underwater robotic vehicles and high performance aircraft.

---

49. A Modular and Hybrid Connectionist System for Speaker Identification

Younès Bennani

https://doi.org/10.1162/neco.1995.7.4.791 | pp. 791-798

This paper presents and evaluates a modular/hybrid connectionist system for speaker identification. Modularity has emerged as a powerful technique for reducing the complexity of connectionist systems, and allowing a priori knowledge to be incorporated into their design. Text-independent speaker identification is an inherently complex task where the amount of training data is often limited. It thus provides an ideal domain to test the validity of the modular/hybrid connectionist approach. To achieve such identification, we develop, in this paper, an architecture based upon the cooperation of several connectionist modules, and a Hidden Markov Model module. When tested on a population of 102 speakers extracted from the DARPA-TIMIT database, perfect identification was obtained.

50. Error Estimation by Series Association for Neural Network Systems

Keehoon Kim, E.B. Bartlett

https://doi.org/10.1162/neco.1995.7.4.799 | pp. 799-808

Estimation of confidence intervals for neural network outputs is important when the uncertainty of a neural network system must be addressed for safety or reliability. This paper presents a new approach for estimating confidence intervals, which can help users validate neural network outputs. The estimation of confidence intervals, called error estimation by series association, is performed by a supplementary neural network trained to predict the error of the main neural network using input features and the output of the main network. The accuracy of this approach is shown using a simple nonlinear mapping and more complicated, realistic nuclear power plant fault diagnosis problems. The results demonstrate that the approach performs confidence estimation successfully.

51. Test Error Fluctuations in Finite Linear Perceptrons

David Barber, David Saad, Peter Sollich

https://doi.org/10.1162/neco.1995.7.4.809 | pp. 809-821

We examine the fluctuations in the test error induced by random, finite, training and test sets for the linear perceptron of input dimension n with a spherically constrained weight vector. This variance enables us to address such issues as the partitioning of a data set into a test and training set. We find that the optimal assignment of the test set size scales with n 2/3 .

52. Learning and Extracting Initial Mealy Automata with a Modular Neural Network Model

Peter Ti■o, Jozef Šajda

https://doi.org/10.1162/neco.1995.7.4.822 | pp. 822-844

A hybrid recurrent neural network is shown to learn small initial mealy machines (that can be thought of as translation machines translating input strings to corresponding output strings, as opposed to recognition automata that classify strings as either grammatical or nongrammatical) from positive training samples. A well-trained neural net is then presented once again with the training set and a Kohonen self-organizing map with the “star” topology of neurons is used to quantize recurrent network state space into distinct regions representing corresponding states of a mealy machine being learned. This enables us to extract the learned mealy machine from the trained recurrent network. One neural network (Kohonen self-organizing map) is used to extract meaningful information from another network (recurrent neural network).

53. Dynamic Cell Structure Learns Perfectly Topology Preserving Map

Jörg Brüske, Gerald Sommer

https://doi.org/10.1162/neco.1995.7.4.845 | pp. 845-865

Dynamic cell structures (DCS) represent a family of artificial neural architectures suited both for unsupervised and supervised learning. They belong to the recently (Martinetz 1994) introduced class of topology representing networks (TRN) that build perfectly topology preserving feature maps. DCS employ a modified Kohonen learning rule in conjunction with competitive Hebbian learning. The Kohonen type learning rule serves to adjust the synaptic weight vectors while Hebbian learning establishes a dynamic lateral connection structure between the units reflecting the topology of the feature manifold. In case of supervised learning, i.e., function approximation, each neural unit implements a radial basis function, and an additional layer of linear output units adjusts according to a delta-rule. DCS is the first RBF-based approximation scheme attempting to concurrently learn and utilize a perfectly topology preserving map for improved performance. Simulations on a selection of CMU-Benchmarks indicate that the DCS idea applied to the growing cell structure algorithm (Fritzke 1993c) leads to an efficient and elegant algorithm that can beat conventional models on similar tasks.

---

54. Methods For Combining Experts' Probability Assessments

Robert A. Jacobs

https://doi.org/10.1162/neco.1995.7.5.867 | pp. 867-888

This article reviews statistical techniques for combining multiple probability distributions. The framework is that of a decision maker who consults several experts regarding some events. The experts express their opinions in the form of probability distributions. The decision maker must aggregate the experts' distributions into a single distribution that can be used for decision making. Two classes of aggregation methods are reviewed. When using a supra Bayesian procedure, the decision maker treats the expert opinions as data that may be combined with its own prior distribution via Bayes' rule. When using a linear opinion pool, the decision maker forms a linear combination of the expert opinions. The major feature that makes the aggregation of expert opinions difficult is the high correlation or dependence that typically occurs among these opinions. A theme of this paper is the need for training procedures that result in experts with relatively independent opinions or for aggregation methods that implicitly or explicitly model the dependence among the experts. Analyses are presented that show that m dependent experts are worth the same as k independent experts where k < or = m. In some cases, an exact value for k can be given; in other cases, lower and upper bounds can be placed on k.

---

55. The Helmholtz Machine

Peter Dayan, Geoffrey E. Hinton, Radford M. Neal, Richard S. Zemel

https://doi.org/10.1162/neco.1995.7.5.889 | pp. 889-904

Discovering the structure inherent in a set of patterns is a fundamental aim of statistical inference or learning. One fruitful approach is to build a parameterized stochastic generative model, independent draws from which are likely to produce the patterns. For all but the simplest generative models, each pattern can be generated in exponentially many ways. It is thus intractable to adjust the parameters to maximize the probability of the observed patterns. We describe a way of finessing this combinatorial explosion by maximizing an easily computed lower bound on the probability of the observations. Our method can be viewed as a form of hierarchical self-supervised learning that may relate to the function of bottom-up and top-down cortical processing pathways.

56. Spontaneous Excitations in the Visual Cortex: Stripes, Spirals, Rings, and Collective Bursts

Corinna Fohlmeister, Wulfram Gerstner, Raphael Ritz, J. Leo van Hemmen

https://doi.org/10.1162/neco.1995.7.5.905 | pp. 905-914

As a simple model of the cortical sheet, we study a locally connected net of spiking neurons. Refractoriness, noise, axonal delays, and the time course of excitatory and inhibitory postsynaptic potentials are taken into account explicitly. In addition to a low-activity state and depending on the synaptic efficacy, four different scenarios evolve spontaneously, viz., stripes, spirals, rings, and collective bursts. Our results can be related to experimental observations of drug-induced epilepsy and hallucinations.

57. Time-Domain Solutions of Oja's Equations

John L. Wyatt, Ibrahim M. Elfadel

https://doi.org/10.1162/neco.1995.7.5.915 | pp. 915-922

Oja's equations describe a well-studied system for unsupervised Hebbian learning of principal components. This paper derives the explicit time-domain solution of Oja's equations for the single-neuron case. It also shows that, under a linear change of coordinates, these equations are a gradient system in the general multi-neuron case. This latter result leads to a new Lyapunov-like function for Oja's equations.

58. Learning the Initial State of a Second-Order Recurrent Neural Network during Regular-Language Inference

Mikel L. Forcada, Rafael C. Carrasco

https://doi.org/10.1162/neco.1995.7.5.923 | pp. 923-930

Recent work has shown that second-order recurrent neural networks (2ORNNs) may be used to infer regular languages. This paper presents a modified version of the real-time recurrent learning (RTRL) algorithm used to train 2ORNNs, that learns the initial state in addition to the weights. The results of this modification, which adds extra flexibility at a negligible cost in time complexity, suggest that it may be used to improve the learning of regular languages when the size of the network is small.

59. An Algebraic Framework to Represent Finite State Machines in Single-Layer Recurrent Neural Networks

René Alquézar, Alberto Sanfeliu

https://doi.org/10.1162/neco.1995.7.5.931 | pp. 931-949

In this paper we present an algebraic framework to represent finite state machines (FSMs) in single-layer recurrent neural networks (SLRNNs), which unifies and generalizes some of the previous proposals. This framework is based on the formulation of both the state transition function and the output function of an FSM as a linear system of equations, and it permits an analytical explanation of the representational capabilities of first-order and higher-order SLRNNs. The framework can be used to insert symbolic knowledge in RNNs

prior to learning from examples and to keep this knowledge while training the network. This approach is valid for a wide range of activation functions, whenever some stability conditions are met. The framework has already been used in practice in a hybrid method for grammatical inference reported elsewhere (Sanfeliu and Alquézar 1994).

---

60. Local and Global Optimization Algorithms for Generalized Learning Automata

V.V. Phansalkar, M.A.L. Thathachar

https://doi.org/10.1162/neco.1995.7.5.950 | pp. 950-973

This paper analyzes the long-term behavior of the REINFORCE and related algorithms (Williams 1986, 1988, 1992) for generalized learning automata (Narendra and Thathachar 1989) for the associative reinforcement learning problem (Barto and Anandan 1985). The learning system considered here is a feedforward connectionist network of generalized learning automata units. We show that REINFORCE is a gradient ascent algorithm but can exhibit unbounded behavior. A modified version of this algorithm, based on constrained optimization techniques, is suggested to overcome this disadvantage. The modified algorithm is shown to exhibit local optimization properties. A global version of the algorithm, based on constant temperature heat bath techniques, is also described and shown to converge to the global maximum. All algorithms are analyzed using weak convergence techniques.

---

61. From Data Distributions to Regularization in Invariant Learning

Todd K. Leen

https://doi.org/10.1162/neco.1995.7.5.974 | pp. 974-981

Ideally pattern recognition machines provide constant output when the inputs are transformed under a group G of desired invariances. These invariances can be achieved by enhancing the training data to include examples of inputs transformed by elements of G, while leaving the corresponding targets unchanged. Alternatively the cost function for training can include a regularization term that penalizes changes in the output when the input is transformed under the group. This paper relates the two approaches, showing precisely the sense in which the regularized cost function approximates the result of adding transformed examples to the training data. We introduce the notion of a probability distribution over the group transformations, and use this to rewrite the cost function for the enhanced training data. Under certain conditions, the new cost function is equivalent to the sum of the original cost function plus a regularizer. For unbiased models, the regularizer reduces to the intuitively obvious choice—a term that penalizes changes in the output when the inputs are transformed under the group. For infinitesimal transformations, the coefficient of the regularization term reduces to the variance of the distortions introduced into the training data. This correspondence provides a simple bridge between the two approaches.

---

62. Initializing Weights of a Multilayer Perceptron Network by Using the Orthogonal Least Squares Algorithm

Mikko Lehtokangas, Jukka Saarinen, Kimmo Kaski, Pentti Huuhtanen

https://doi.org/10.1162/neco.1995.7.5.982 | pp. 982-999

Usually the training of a multilayer perceptron network starts by initializing the network weights with small random values, and then the weight adjustment is carried out by using an iterative gradient descent-based optimization routine called backpropagation training. If the random initial weights happen to be far from a good solution or they are near a poor local optimum, the training will take a lot of time since many iteration steps are required. Furthermore, it is very possible that the network will not converge to an adequate solution at all. On the other hand, if the initial weights are close to a good solution the training will be much faster and the possibility of obtaining adequate convergence increases. In this paper a new method for initializing the weights is presented. The method is based on the orthogonal least squares algorithm. The simulation results obtained with the proposed initialization method show a considerable improvement in training compared to the randomly initialized networks. In light of practical experiments, the proposed method has proven to be fast and useful for initializing the network weights.

---

63. Learning and Generalization in Radial Basis Function Networks

Jason A. S. Freeman, David Saad

https://doi.org/10.1162/neco.1995.7.5.1000 | pp. 1000-1020

The two-layer radial basis function network, with fixed centers of the basis functions, is analyzed within a stochastic training paradigm. Various definitions of generalization error are considered, and two such definitions are employed in deriving generic learning curves and generalization properties, both with and without a weight decay term. The generalization error is shown analytically to be related to the evidence and, via the evidence, to the prediction error and free energy. The generalization behavior is explored; the generic learning curve is found to be inversely proportional to the number of training pairs presented. Optimization of training is considered by minimizing the generalization error with respect to the free parameters of the training algorithms. Finally, the effect of the joint activations between hidden-layer units is examined and shown to speed training.

---

64. Precision and Approximate Flatness in Artificial Neural Networks

Maxwell B. Stinchcombe

https://doi.org/10.1162/neco.1995.7.5.1021 | pp. 1021-1039

Several of the major classes of artificial neural networks' output functions are linear combinations of elements of approximately flat sets. This gives a tool for understanding the precision problem as well as providing a rationale for mixing types of networks. Approximate flatness also helps explain the power of artificial neural network techniques relative to series regressions—series regressions take linear combinations of flat sets, while neural networks take linear combinations of the much larger class of approximately flat sets.

---

65. Lower Bounds on the VC Dimension of Smoothly Parameterized Function Classes

Wee Sun Lee, Peter L. Bartlett, Robert C. Williamson

https://doi.org/10.1162/neco.1995.7.5.1040 | pp. 1040-1053

We examine the relationship between the VC dimension and the number of parameters of a threshold smoothly parameterized function class. We show that the VC dimension of such a function class is at least k if there exists a k-dimensional differentiable manifold in the parameter space such that each member of the manifold corresponds to a different decision boundary. Using this result, we are able to obtain lower bounds on the VC dimension proportional to the number of parameters for several thresholded function classes including two-layer neural networks with certain smooth activation functions and radial basis functions with a gaussian basis. These lower bounds hold even if the magnitudes of the parameters are restricted to be arbitrarily small. In Valiant's probably approximately correct learning framework, this implies that the number of examples necessary for learning these function classes is at least linear in the number of parameters.

---

66. Agnostic PAC Learning of Functions on Analog Neural Nets

Wolfgang Maass

https://doi.org/10.1162/neco.1995.7.5.1054 | pp. 1054-1078

We consider learning on multilayer neural nets with piecewise polynomial activation functions and a fixed number k of numerical inputs. We exhibit arbitrarily large network architectures for which efficient and provably successful learning algorithms exist in the rather realistic refinement of Valiant's model for probably approximately correct learning ("PAC learning") where no a priori assumptions are required about the "target function" (agnostic learning), arbitrary noise is permitted in the training sample, and the target outputs as well as the network outputs may be arbitrary reals. The number of computation steps of the learning algorithm LEARN that we construct is bounded by a polynomial in the bit-length n of the fixed number of input variables, in the bound s for the allowed bit-length of weights, in $1/\varepsilon$, where ε is some arbitrary given bound for the true error of the neural net after training, and in $1/\delta$ where δ is some arbitrary given bound for the probability that the learning algorithm fails for a randomly drawn training sample. However, the computation time of LEARN is exponential in the number of weights of the considered network architecture, and therefore only of interest for neural nets of small size. This article provides details to the previously published extended abstract (Maass 1994).

---

67. Convex Potentials and their Conjugates in Analog Mean-Field Optimization

Ibrahim M. Elfadel

https://doi.org/10.1162/neco.1995.7.5.1079 | pp. 1079-1104

This paper deals with the problem of mapping hybrid (i.e., both discrete and continuous) constrained optimization problems onto analog networks. The saddle-point paradigm of mean-field methods in statistical physics provides a systematic procedure for finding such a mapping via the notion of effective energy. Specifically, it is shown that within this paradigm, to each closed bounded constraint set is associated a smooth convex potential function. Using the conjugate (or the Legendre-Fenchel transform) of the convex potential, the effective energy can be transformed to yield a cost function that is a natural generalization of the analog Hopfield energy. Descent dynamics and deterministic annealing can then be used to find the global minimum of the original minimization problem. When the conjugate is hard to compute explicitly, it is

shown that a minimax dynamics, similar to that of Arrow and Hurwicz in Lagrangian optimization, can be used to find the saddle points of the effective energy. As an illustration of its wide applicability, the effective energy framework is used to derive Hopfield-like energy functions and descent dynamics for two classes of networks previously considered in the literature, winner-take-all networks and rotor networks, even when the cost function of the original optimization problem is not quadratic.

---

68. Patterns of Functional Damage in Neural Network Models of Associative Memory

Eytan Ruppin, James A. Reggia

https://doi.org/10.1162/neco.1995.7.5.1105 | pp. 1105-1127

Current understanding of the effects of damage on neural networks is rudimentary, even though such understanding could lead to important insights concerning neurological and psychiatric disorders. Motivated by this consideration, we present a simple analytical framework for estimating the functional damage resulting from focal structural lesions to a neural network model. The effects of focal lesions of varying area, shape, and number on the retrieval capacities of a spatially organized associative memory are quantified, leading to specific scaling laws that may be further examined experimentally. It is predicted that multiple focal lesions will impair performance more than a single lesion of the same size, that slit like lesions are more damaging than rounder lesions, and that the same fraction of damage (relative to the total network size) will result in significantly less performance decrease in larger networks. Our study is clinically motivated by the observation that in multi-infarct dementia, the size of metabolically impaired tissue correlates with the level of cognitive impairment more than the size of structural damage. Our results account for the detrimental effect of the number of infarcts rather than their overall size or structural damage, and for the "multiplicative" interaction between Alzheimer's disease and multi-infarct dementia.

---

69. An Information-Maximization Approach to Blind Separation and Blind Deconvolution

Anthony J. Bell, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1995.7.6.1129 | pp. 1129-1159

We derive a new self-organizing learning algorithm that maximizes the information transferred in a network of nonlinear units. The algorithm does not assume any knowledge of the input distributions, and is defined here for the zero-noise limit. Under these conditions, information maximization has extra properties not found in the linear case (Linsker 1989). The nonlinearities in the transfer function are able to pick up higher-order moments of the input distributions and perform something akin to true redundancy reduction between units in the output representation. This enables the network to separate statistically independent components in the inputs: a higher-order generalization of principal components analysis. We apply the network to the source separation (or cocktail party) problem, successfully separating unknown mixtures of up to 10 speakers. We also show that a variant on the network architecture is able to perform blind deconvolution (cancellation of unknown echoes and reverberation in a speech signal). Finally, we derive dependencies of information transfer on time delays. We suggest that information maximization provides a unifying framework for problems in "blind" signal processing.

---

70. A Perceptron Reveals the Face of Sex

Michael S. Gray, David T. Lawrence, Beatrice A. Golomb, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1995.7.6.1160 | pp. 1160-1164

ermine how the reliability of sex discrimination is related to resolution. A normalized pixel-based representation was used for the faces because it explicitly retained texture and shape information while also maintaining geometric relationships. We found that the linear perceptron model can classify sex from facial images with 81% accuracy, compared to 92% accuracy with compression coding on the same data set [6]. The advantage of using a simple linear perceptron with normalized pixel-based inputs is that it allows us to see explicitly those regions of the face that make the largest and most reliable contributions to the classification of sex. A database of 90 faces (44 males, 46 females) was used (O'Toole, Millward, & Anderson [9]). No facial hair, jewelry, or makeup was on any of the faces. Each face was rotated until the eyes were level, and then scaled and cropped so that each image showed a similar facial area. From the original set of faces, we created 5 separate databases at 5

---

71. Self-Organization as an Iterative Kernel Smoothing Process

Filip Mulier, Vladimir Cherkassky

https://doi.org/10.1162/neco.1995.7.6.1165 | pp. 1165-1177

Kohonen's self-organizing map, when described in a batch processing mode, can be interpreted as a statistical kernel smoothing problem. The batch SOM algorithm consists of two steps. First, the training data are partitioned according to the Voronoi regions of the map unit locations. Second, the units are updated by taking weighted centroids of the data falling into the Voronoi regions, with the weighing function given by the neighborhood. Then, the neighborhood width is decreased and steps 1, 2 are repeated. The second step can be interpreted as a statistical kernel smoothing problem where the neighborhood function corresponds to the kernel and neighborhood width corresponds to kernel span. To determine the new unit locations, kernel smoothing is applied to the centroids of the Voronoi regions in the topological space. This interpretation leads to some new insights concerning the role of the neighborhood and dimensionality reduction. It also strengthens the algorithm's connection with the Principal Curve algorithm. A generalized self-organizing algorithm is proposed, where the kernel smoothing step is replaced with an arbitrary nonparametric regression method.

---

72. On the Distribution and Convergence of Feature Space in Self-Organizing Maps

Hujun Yin, N.M. Allinson

https://doi.org/10.1162/neco.1995.7.6.1178 | pp. 1178-1187

In this paper an analysis of the statistical and the convergence properties of Kohonen's self-organizing map of any dimension is presented. Every feature in the map is considered as a sum of a number of random variables. We extend the Central Limit Theorem to a particular case, which is then applied to prove that the feature space during learning tends to multiple gaussian distributed stochastic processes, which will eventually converge in the mean-square sense to the probabilistic centers of input subsets to form a quantization mapping with a minimum mean squared distortion either globally or locally. The diminishing

effect, as training progresses, of the initial states on the value of the feature map is also shown.

---

73. Sorting with Self-Organizing Maps

Marco Budinich

https://doi.org/10.1162/neco.1995.7.6.1188 | pp. 1188-1190

A self-organizing feature map (Von der Malsburg 1973; Kohonen 1984) sorts n real numbers in O(n) time apparently violating the O(n log n) bound. Detailed analysis shows that the net takes advantage of the uniform distribution of the numbers and, in this case, sorting in O(n) is possible. There are, however, an exponentially small fraction of pathological distributions producing O(n2) sorting time. It is interesting to observe that standard learning produced a smart sorting algorithm.

---

74. Introducing Asymmetry into Interneuron Learning

Colin Fyfe

https://doi.org/10.1162/neco.1995.7.6.1191 | pp. 1191-1205

A review is given of a new artificial neural network architecture in which the weights converge to the principal component subspace. The weights learn by only simple Hebbian learning yet require no clipping, normalization or weight decay. The net self-organizes using negative feedback of activation from a set of "interneurons" to the input neurons. By allowing this negative feedback from the interneurons to act on other interneurons we can introduce the necessary asymmetry to cause convergence to the actual principal components. Simulations and analysis confirm such convergence.

---

75. Learning and Generalization with Minimerror, A Temperature-Dependent Learning Algorithm

Bruno Raffin, Mirta B. Gordon

https://doi.org/10.1162/neco.1995.7.6.1206 | pp. 1206-1224

We study the numerical performances of Minimerror, a recently introduced learning algorithm for the perceptron that has analytically been shown to be optimal both on learning linearly and nonlinearly separable functions. We present its implementation on learning linearly separable boolean functions. Numerical results are in excellent agreement with the theoretical predictions.

---

76. Regularized Neural Networks: Some Convergence Rate Results

Valentina Corradi, Halbert White

https://doi.org/10.1162/neco.1995.7.6.1225 | pp. 1225-1244

In a recent paper, Poggio and Girosi (1990) proposed a class of neural networks obtained from the theory of regularization. Regularized networks are capable of approximating arbitrarily well any continuous function on a compactum. In this paper we consider in detail the learning problem for the one-dimensional case. We show that in the case of output data observed with noise, regularized networks are capable of learning and

approximating (on compacta) elements of certain classes of Sobolev spaces, known as reproducing kernel Hilbert spaces (RKHS), at a nonparametric rate that optimally exploits the smoothness properties of the unknown mapping. In particular we show that the total squared error, given by the sum of the squared bias and the variance, will approach zero at a rate of n(-2m)/(2m+1), where m denotes the order of differentiability of the true unknown function. On the other hand, if the unknown mapping is a continuous function but does not belong to an RKHS, then there still exists a unique regularized solution, but this is no longer guaranteed to converge in mean square to a well-defined limit. Further, even if such a solution converges, the total squared error is bounded away from zero for all n sufficiently large.

---

77. The Target Switch Algorithm: A Constructive Learning Procedure for Feed-Forward Neural Networks

Colin Campbell, Consuelo Vicente

https://doi.org/10.1162/neco.1995.7.6.1245 | pp. 1245-1264

We propose an efficient procedure for constructing and training a feed-forward neural network. The network can perform binary classification for binary or analogue input data. We show that the procedure can also be used to construct feedforward neural networks with binary-valued weights. Neural networks with binary-valued weights are potentially straightforward to implement using microelectronic or optical devices and they can also exhibit good generalization.

---

78. On the Practical Applicability of VC Dimension Bounds

Sean B. Holden, Mahesan Niranjan

https://doi.org/10.1162/neco.1995.7.6.1265 | pp. 1265-1288

This article addresses the question of whether some recent Vapnik-Chervonenkis (VC) dimension-based bounds on sample complexity can be regarded as a practical design tool. Specifically, we are interested in bounds on the sample complexity for the problem of training a pattern classifier such that we can expect it to perform valid generalization. Early results using the VC dimension, while being extremely powerful, suffered from the fact that their sample complexity predictions were rather impractical. More recent results have begun to improve the situation by attempting to take specific account of the precise algorithm used to train the classifier. We perform a series of experiments based on a task involving the classification of sets of vowel formant frequencies. The results of these experiments indicate that the more recent theories provide sample complexity predictions that are significantly more applicable in practice than those provided by earlier theories; however, we also find that the recent theories still have significant shortcomings.

---

79. LeRec: A NN/HMM Hybrid for On-Line Handwriting Recognition

Yoshua Bengio, Yann LeCun, C.R. Nohl, Chris Burges

https://doi.org/10.1162/neco.1995.7.6.1289 | pp. 1289-1303

We introduce a new approach for on-line recognition of handwritten words written in unconstrained mixed style. The preprocessor performs a word-level normalization by fitting a model of the word structure using the EM algorithm. Words are then coded into low resolution "annotated images" where each pixel contains

information about trajectory direction and curvature. The recognizer is a convolution network that can be spatially replicated. From the network output, a hidden Markov model produces word scores. The entire system is globally trained to minimize word-level errors.

---

80. Index

(authors unknown)

https://doi.org/10.1162/neco.1995.7.6.1305 | pp. 1305-1314

No abstract available.